Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Topography-aware Temperature Forecasting in Europe

Jan Zgraggen[1], Chang Xu[1*] and Devis Tuia[1]

[1]EPFL, Switzerland
*Corresponding author. Email: chang.xu@epfl.ch

Received 23 March 2026

Keywords: weather forecasting, deep learning, physics-informed neural networks

1. Introduction and objectives

The increasing frequency of occurrence of heatwaves has increased the importance of accurate temperature forecasting at the regional to continental scale. In weather forecasting, machine learning (ML) approaches have demonstrated performance matching that of numerical methods, while providing higher efficiency and inference speed. However, most data-driven models do not explicitly account for physical relationships [1]. Atmospheric variables (e.g., temperature) are governed by physical constraints such as terrain variability, and these constraints are especially pronounced in complex regions such as the Alps, leading to spatially varying biases, which have been observed in the forecasts [2]. Incorporating physical priors into ML models has thus become an important research objective, but so far only few studies have explored how to effectively incorporate terrain information into temperature forecasting. In this paper, we investigate how to leverage terrain priors for topography-aware temperature forecasting across Europe from three perspectives: fusion, location embeddings, and loss regularization (Figure 2). Results on the CERRA dataset show consistent improvements related to the three strategies, with the greatest improvement obtained by location embeddings. In particular, the proposed regularization framework demonstrates significant improvements over strong embedding-based methods and yields promising results for geography-aware forecasting with ML.

2. Methods and data

Simple Fusion. A simple way to incorporate physical information is to add it as an extra input channel. We concatenate the temperature input $\mathbf{X} \in \mathbb{R}^{T\times C_{var}\times H\times W}$ with the terrain information at the channel dimension, resulting in an input $\mathbf{X}_{in} \in \mathbb{R}^{T\times(C_{var}+1)\times H\times W}$. To enhance data interaction, we also perform element-wise multiplication of the original temperature channel and terrain information.

Geographical Implicit Neural Representation (GINR). GINR learns a positional encoding $\mathbf{PE}(\mathbf{G}) : \mathbb{R}^{2\times H\times W} \rightarrow \mathbb{R}^{N_{coeff}\times H\times W}$ for geography $\mathbf{G}(\lambda,\phi)$ (accounting for both location $\mathbf{Y}(\lambda,\phi)$ and elevation $\mathbf{H}(\lambda,\phi)$). We employ GeoFAR [3] by projecting the spherical harmonics (SH)-encoded locations $\mathbf{Y_{sh}}$ and terrain embeddings

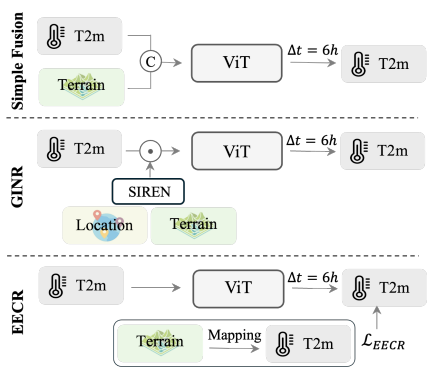

Figure 1. *Different ways of topography-aware temperature forecasting.*

© The Authors(s) 2020. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted re-use, distribution, and reproduction in any medium, provided the original work is properly cited.

$\mathbf{T} = (\mathbf{H}, \partial_\lambda \mathbf{H}, \partial_\phi \mathbf{H})$ into the latent space with the Sinusoidal Representation Network (SIREN) [4]: $\mathbf{PE}(\mathbf{G}) = \mathrm{NN}_{\mathrm{SIREN}}\big(\mathbf{Y_{sh}} + \mathrm{MLP}(\mathbf{T})\big)$. We then project the temperature input $\mathbf{X}$ to the same dimension as the **PE**, then perform a per-pixel multiplication between temperature and **PE** as the model input.

Explicit Empirical Consistency Regularization (EECR). EECR is a regularization framework that enforces statistical consistency between model predictions and empirically observed relationships across physical domains. We use EECR to constrain the predicted temperature fields to be consistent with terrain–temperature relationships learned from training data. EECR has two components: a *Spatial Operator*,$\phi_D : \mathcal{T}_D^{in} \to \mathcal{T}_D^{target}$, which aggregates or extrapolates spatial information while preserving the physical units of domain D; and a *Domain Mapping*, $\psi_{D\to G} : \mathcal{T}_D^{\alpha} \to \mathcal{T}_G^{\alpha}$, which maps quantities between physical domains while acting point-wise over input grids α. EECR then aligns aggregated statistics of a prior p in domain D with corresponding aggregated statistics of the predictions $\hat{y}$ in domain G by minimizing a consistency loss between $\psi_{D\to G}(\phi_D(p))$ and $\phi_G(\hat{y})$. We introduce three approaches for instantiating EECR to capture the *dataset-specific* relationship between terrain gradients and the resulting temperature differences (Table 1).

Histogram (HIST): We compute local orographic differences ($|\Delta H|$), bin them into $\{\mathcal{B}_k\}_{k=1}^{K}$, and estimate the expected absolute temperature difference per bin: $\hat{g}(b_k) = \mathbb{E}\left[|\Delta T| \;\middle|\; |\Delta H| \in \mathcal{B}_k\right]$. To obtain a continous domain mapping, $\hat{g}$ is smoothed using the smoothing function $\mathcal{S} = \hat{\sigma}((0.5H + 15.6)10^{-3}), \hat{\sigma} = \sigma(\frac{x-130}{20})$ where the sigmoid function σ is used for weighting. *Space-Dependent* (SD): since the topography–temperature relationship varies across locations, we use a parametric mapping to learn it, while remaining translation-invariant. *Geography & Space-Dependent* (GSD): Following the GINR framework above, one can constrain the learned mapping space by using a GINR block with Convolution head as domain mapping.

Table 1. *EECR instanciations.*

Method	ϕ_H	ϕ_T	$\psi^{\boldsymbol{\theta}}_{H\to T}$
HIST	$\lvert\widehat{\nabla}\rvert(H)$	$\lvert\widehat{\nabla}\rvert(T)$	$\mathcal{S}(\hat{g}(b_k))(\cdot)$
SD	$\lvert\widehat{\nabla}\rvert(H)$	$\lvert\widehat{\nabla}\rvert(T)$	$\mathrm{Conv2D}_{3\times3}(\cdot)$
GSD	H	$\lvert\widehat{\nabla}\rvert(T)$	$\mathrm{Conv2D}_{3\times3}(\mathbf{PE_{GINR}}(\cdot))$

Data. We use the CERRA dataset with 2-m temperature (T2m) and forecast 6 hours in advance. Data are processed to 11 km resolution (534×534) with 3-hourly samples from 2018–2021 (2920 frames/year) [5]. We train on 2018–2020 (8760 frames) and test on 2021 (2920 frames).

3. Preliminary results

We compare the three topography-aware methods based on a ViT [1] (Table 2). All topography-aware approaches improve over the ViT baseline, with a reduction in RMSE up to 10%. The most substantial improvements are observed for geography encoding (GINR+EECR). Among the proposed constraining methods within EECR, HIST performs best, showing the interest of space-agnostic consistency regularization in climate variables forecasting.

Table 2. *A comparison of results across models.*

Model (ViT)	**EECR**	**RMSE ↓**	**MAB ↓**
Baseline	-	2.7910	1.982
Simple Fusion	-	2.7507	1.980
Baseline	HIST	2.7415	1.980
GINR	-	2.5028	1.776
GINR	SD	2.4795	1.761
GINR	GSD	2.4718	1.756
GINR	HIST	**2.4662**	**1.749**

References

[1] T. Nguyen, J. Jewik, H. Bansal, P. Sharma, and A. Grover, "ClimateLearn: Benchmarking machine learning for weather and climate modeling," in *Advances in Neural Information Processing Systems*, 2023, datasets and Benchmarks Track.

[2] T. Haiden, I. Sandu, G. Balsamo, G. Arduini, and A. Beljaars, "Addressing biases in near-surface forecasts," *ECMWF Newsletter*, no. 157, 2018, autumn 2018.

[3] C. Xu, G. Sumbul, L. Mi, R. Zbinden, and D. Tuia, "GeoFAR: Geography-informed frequency-aware super-resolution for climate data," in *The Fourteenth International Conference on Learning Representations*, 2026.

[4] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in *Advances in Neural Information Processing Systems*, 2020.

[5] M. Ridal, E. Bazile, P. Le Moigne *et al.*, "CERRA, the copernicus european regional reanalysis system," *Quarterly Journal of the Royal Meteorological Society*, vol. 150, no. 763, pp. 3385–3411, 2024.